

TRACE THE ACE

BASELINE DEVELOPMENT ROADMAP

Main Dataset Only



Team SkyMap

- 00:00:12 Tutor: What do you think is the first step here?
- 00:00:18 Student: I think we need to identify the variables.
- 00:00:24 Tutor: Good—can you explain why?
- ...





Problem

Predict whether a student will answer the follow-up assessment correctly.



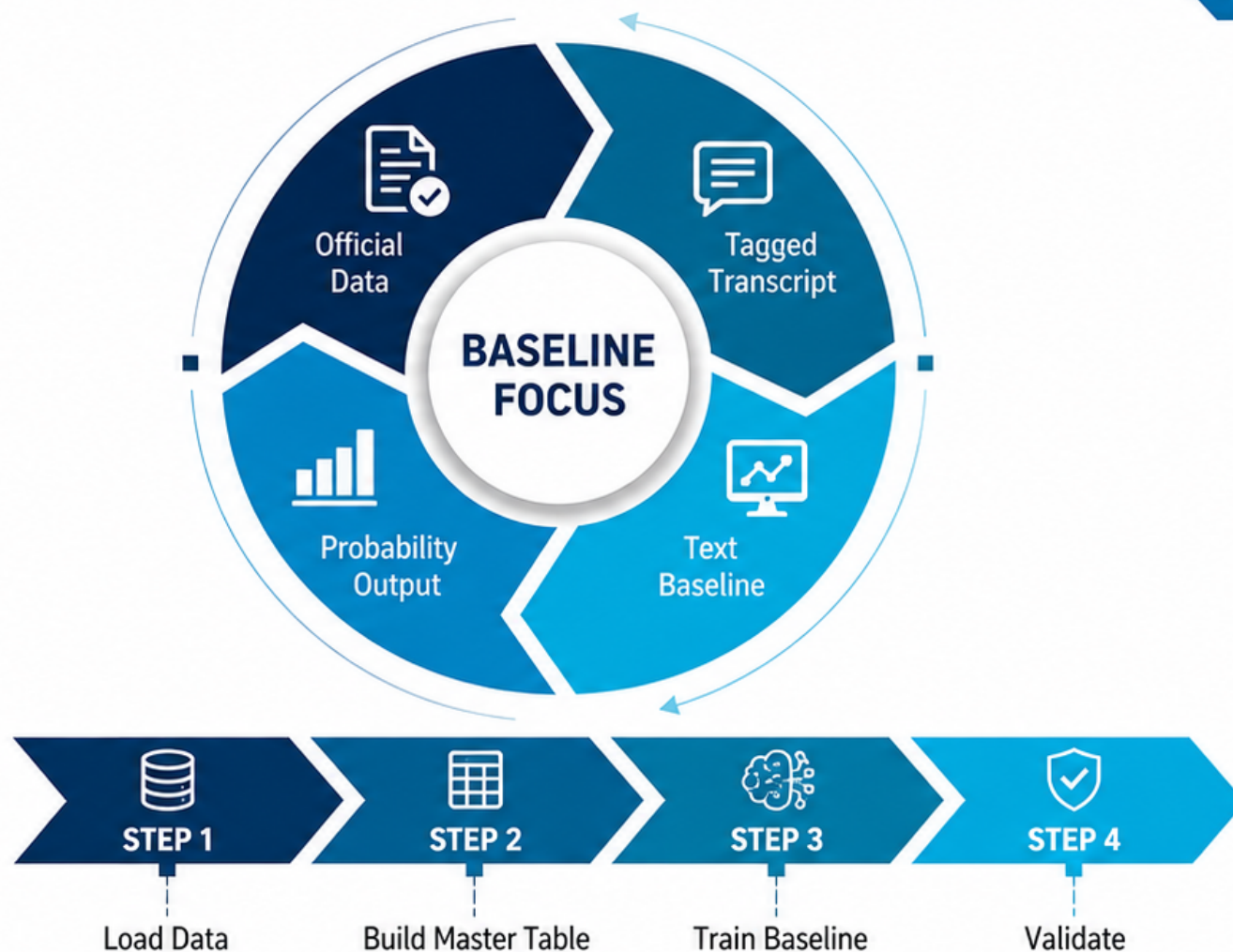
Baseline Scope

Use only the official main dataset: train_features, train_labels and session transcripts.



Baseline Direction

Build a clean response-level master table, train simple reproducible text baselines, and validate by session-level split.



External datasets are excluded from the initial baseline.

DATASET ARCHITECTURE

A



FEATURE FILES

train_features.csv / test_features.csv

- ▶ response_id
- ▶ session_id
- ▶ learning_objective

B



TRANSCRIPT FILES

train_transcripts / test_transcripts

- ▶ utterance_id
- ▶ role
- ▶ content
- ▶ timestamp

C



TRAINING LABELS

train_labels.csv

- ▶ correct = 0 or 1



One transcript may support multiple response rows.

MASTER DATA PREPARATION

Transform raw files into a clean, response-level master table.



RAW FILES

Official train/test datasets



TAGGED TRANSCRIPT

Session transcript with [TUTOR] and [STUDENT] tags



MASTER TABLE

One row per response with label and context

01

Load official train and test files



02

Rebuild each session transcript



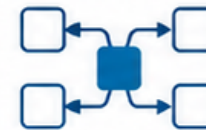
03

Add [TUTOR] and [STUDENT] tags



04

Merge by session_id



05

Merge labels by response_id



EXAMPLE TRANSCRIPT

[TUTOR] What is the first step?
[STUDENT] I think we identify the variables.
[TUTOR] Good—explain why.
[STUDENT] Because they determine what we analyze.



OUTPUT



master_train / master_test

One response-level row =
learning objective + session transcript + label.



BASELINE INPUT DESIGN

The baseline model uses a focused set of inputs from the official dataset to ensure reproducibility and a clean evaluation of core predictive signals.



PURPOSEFUL INPUTS

Each input type captures a distinct aspect of the tutoring interaction relevant to prediction.



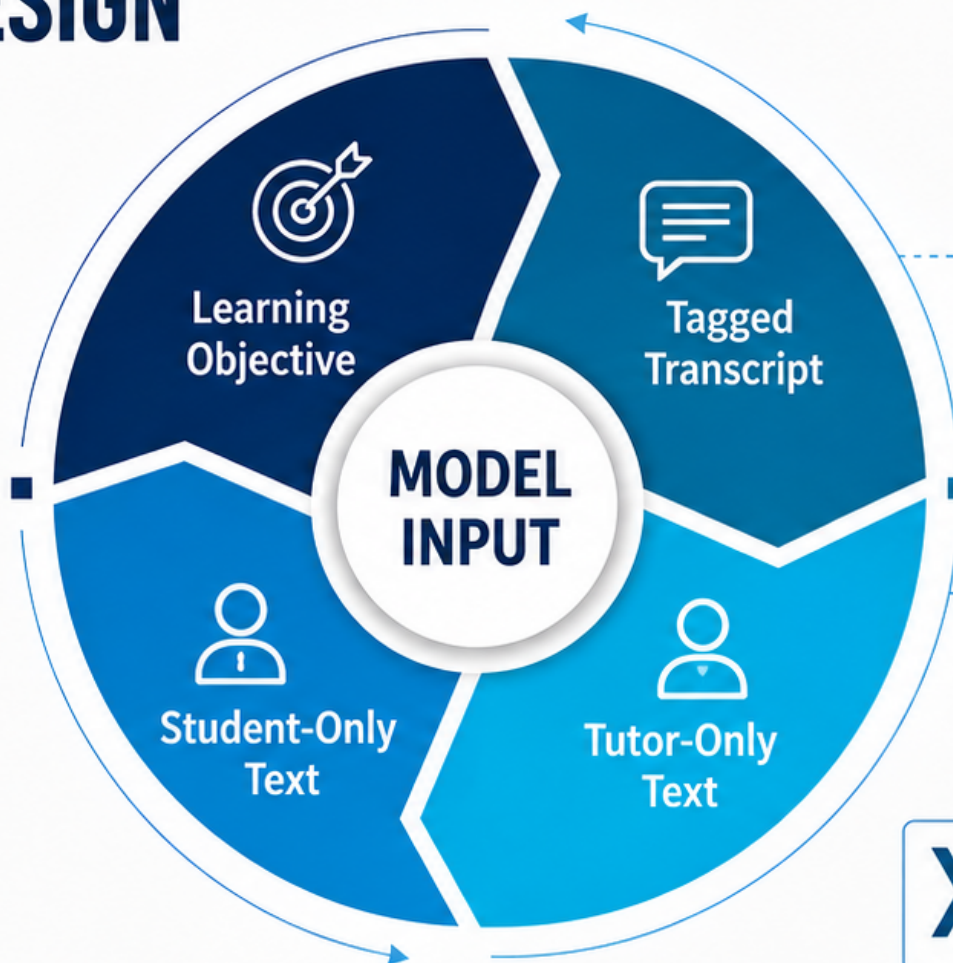
MINIMAL & TRANSPARENT

We restrict inputs to simple, well-defined signals that are easy to understand and replicate.



REPRODUCIBLE BASELINE

Using only the official dataset and simple features establishes a strong, fair baseline for comparison.



TEXT SIGNALS

- words & phrases
- spelling variation
- role tags



SESSION SIGNALS

- turn counts
- word counts
- question counts



Primary baseline input =
learning objective + tagged transcript



Only simple features from the official dataset are used in the baseline.

VALIDATION FRAMEWORK

A LEAKAGE-FREE, SESSION-AWARE PROCESS



1  SPLIT

2  TRAIN

3  PREDICT

4  EVALUATE



TRAINING
SESSIONS



VALIDATION
SESSIONS



TEST
SESSIONS



SPLIT RULE

- ▶ group by `session_id`
- ▶ keep each transcript in one split
- ▶ start with grouped 80/20 split
- ▶ later confirm with GroupKFold



EVALUATION

- ▶ **primary metric:** Log Loss
- ▶ **support metric:** ROC-AUC
- ▶ **benchmark:** constant probability baseline



Random response-level splitting may create transcript leakage.

BASELINE MODEL DEVELOPMENT

A progressive modeling approach that builds from simple to sophisticated to capture complementary language signals.



MODEL 0

Constant
Probability



MODEL 1

Word TF-IDF +
Logistic Regression



MODEL 2

Character TF-IDF +
Logistic Regression



FINAL

Weighted
Ensemble



WORD MODEL

Captures meaningful
words and phrases.



CHARACTER MODEL

Handles spelling and
text variation.



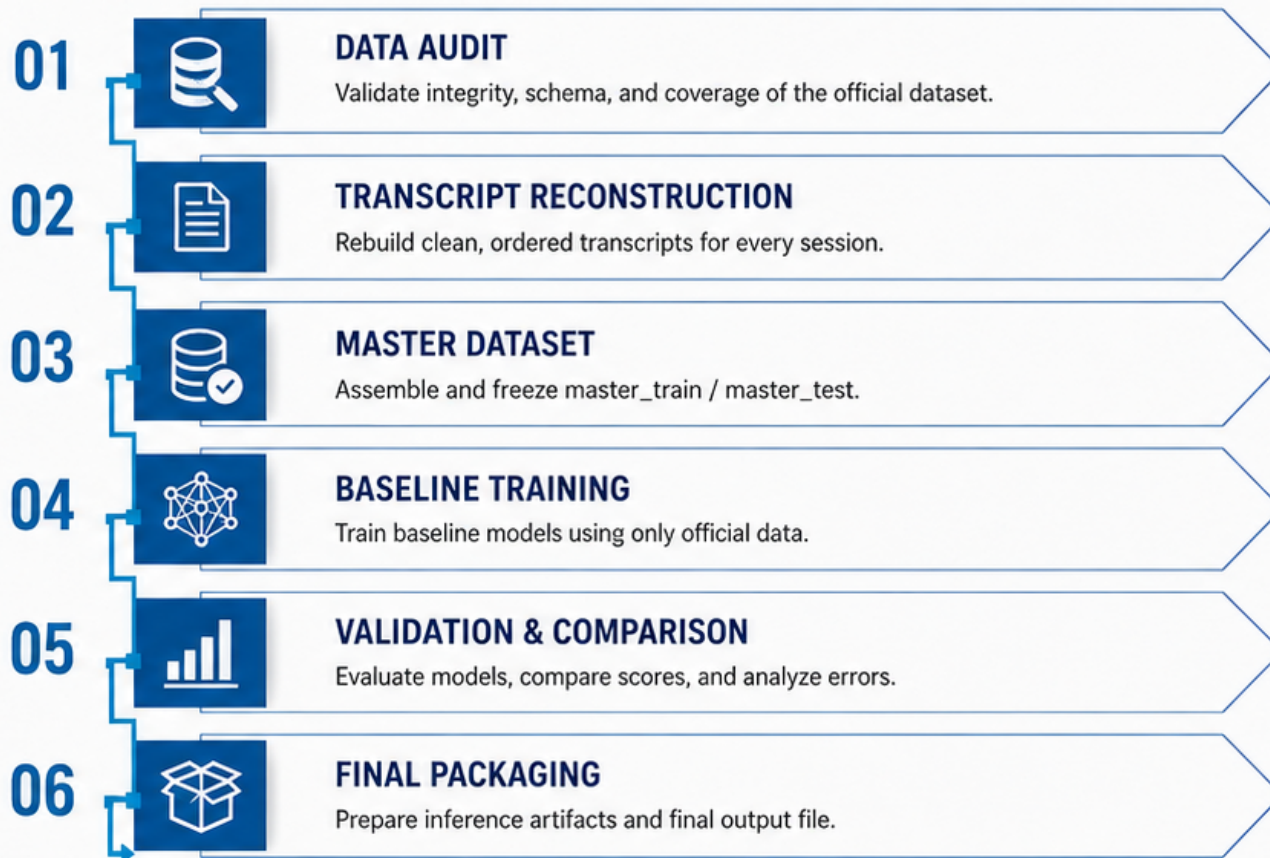
FINAL OUTPUT

Produces a correctness
probability.



Calibration may be added after baseline stability is confirmed.

EXECUTION ROADMAP & DELIVERABLES



DELIVERABLES



DATA

- master_train / master_test



MODELS

- constant baseline
- word TF-IDF
- character TF-IDF
- ensemble



REPORTING

- validation scores
- error analysis



SUBMISSION

- inference assets
- output file

